

Java Basics - Notes & Setup Guide

Hi everyone,

These are the notes that I will be explaining in class. Please read them once before coming to the session. Don't worry if you don't understand everything now - we will discuss each topic slowly in class.

You are not expected to memorize or understand everything now. This is only to get familiar with the terms. I will explain each concept step by step in class.

1. What is a Java Program?

A Java program is a set of instructions written in Java language to tell the computer what to do. In class, I explained this using a real-life example like giving step-by-step instructions to make tea.

2. Source Code

The code that we write as programmers is called **source code**. In Java, source code files end with the extension **.java**.

3. How a Java Program Works

In our system, a Java program runs in the following steps:

.java file → Compiler → .class file (bytecode) → JVM → Output

I will explain this flow clearly in class with real-life examples so that you can remember it easily.

4. Compiler

A compiler checks whether our Java code has any syntax errors and converts it into bytecode. You can think of the compiler like a language translator.

5. Bytecode

Bytecode is an intermediate form of code. It is not machine-specific. The JVM understands and runs this bytecode.

6. JVM (Java Virtual Machine)

JVM is responsible for running Java programs. It also manages memory and provides security. In class, I will explain JVM using simple real-life examples.

7. JRE (Java Runtime Environment)

JRE contains the JVM and required libraries to run Java programs. If someone only wants to run a Java program, JRE is enough.

8. JDK (Java Development Kit)

JDK contains JRE, compiler, and other development tools. Since we are learning Java programming, I will ask you to install JDK.

9. Java 21 (LTS) - Direct Download Link

Please download **Java JDK 21 (LTS)** only using the official link below:

<https://www.oracle.com/java/technologies/downloads/#java21>

- Open the link above.
- Select your operating system (Windows or macOS).
- Download the installer (Windows x64 Installer / macOS Installer).
- Run the installer and follow default steps.
- After installation, open Command Prompt or Terminal.
- Type: **java -version**
- If version 21 is shown, Java is installed successfully.

10. VS Code - Direct Download Link

Please download **Visual Studio Code** using the official link below:

<https://code.visualstudio.com/download>

- Open the link above.
- Download VS Code for your operating system.
- Install using default settings.
- Open VS Code after installation.
- Go to the **Extensions** section.
- Search and install: **Extension Pack for Java**.
- Restart VS Code.

Important: If you already have VS Code working with Java code, that's perfectly fine - no need to reinstall. If you face any problems (Java/VS Code not working), please uninstall and download again based on the steps above. If you face any installation issues, do not panic. I will help you step-by-step in the class.